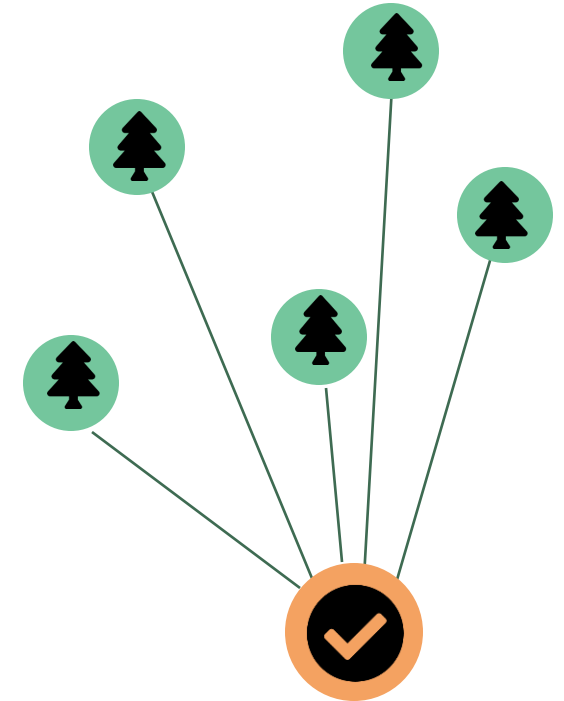


ENSEMBLE LEARNING

Combining Multiple Models for Stronger Predictions



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What is Ensemble Learning?

A technique that combines predictions from multiple models to produce one stronger, more reliable result.



Multiple Learners

Several models ("weak learners") are trained instead of relying on just one



Combined Decision

Their individual outputs are merged by voting or averaging



Lower Error

Combining reduces both bias and variance compared to a single model



Better Accuracy

The combined model typically outperforms any individual member

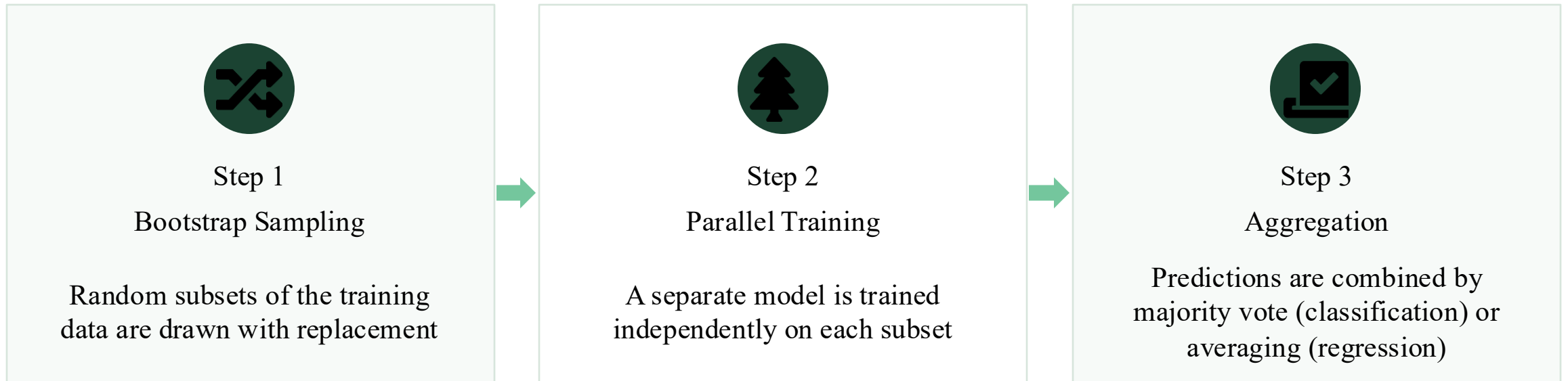


Wisdom of the Crowd

Just as a group's average guess often beats any one expert, a group of models often predicts better than a single model.

Bagging (Bootstrap Aggregating)

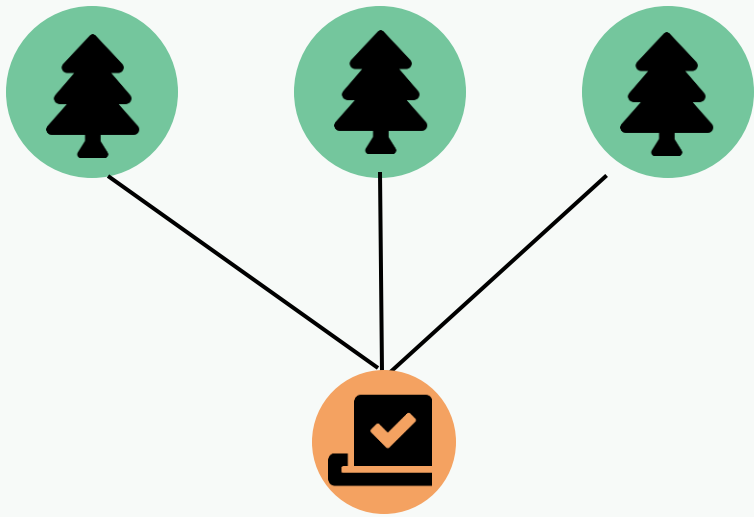
Trains many models in parallel on random subsets of data, then averages their results.



Key benefit: reduces variance and helps prevent overfitting, since each model sees slightly different data.

Random Forest: Bagging in Action

The most widely used bagging algorithm — an ensemble of decision trees.

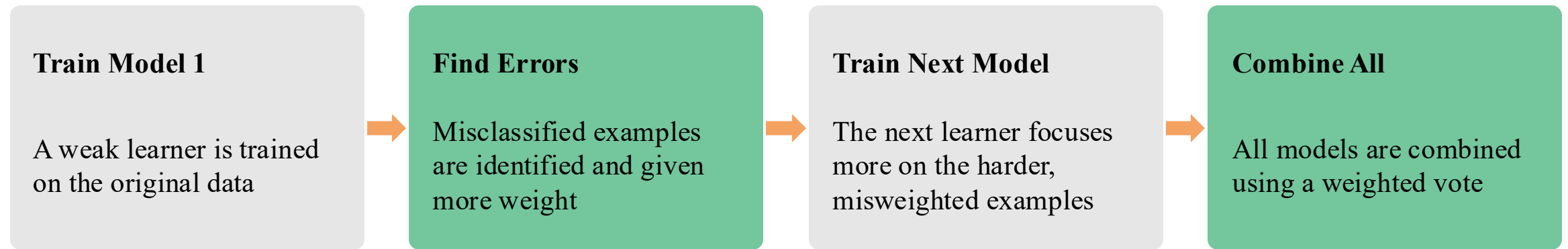


Many decision trees vote on the final result

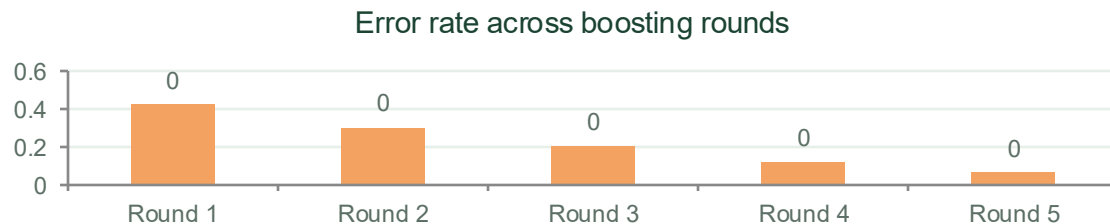
- 1 Builds a large number of decision trees, each on a bootstrapped sample
- 2 At each split, only a random subset of features is considered
- 3 This extra randomness keeps trees diverse and less correlated
- 4 Final prediction: majority vote for classification, average for regression

Boosting

Trains models sequentially, where each new model focuses on the previous one's mistakes.

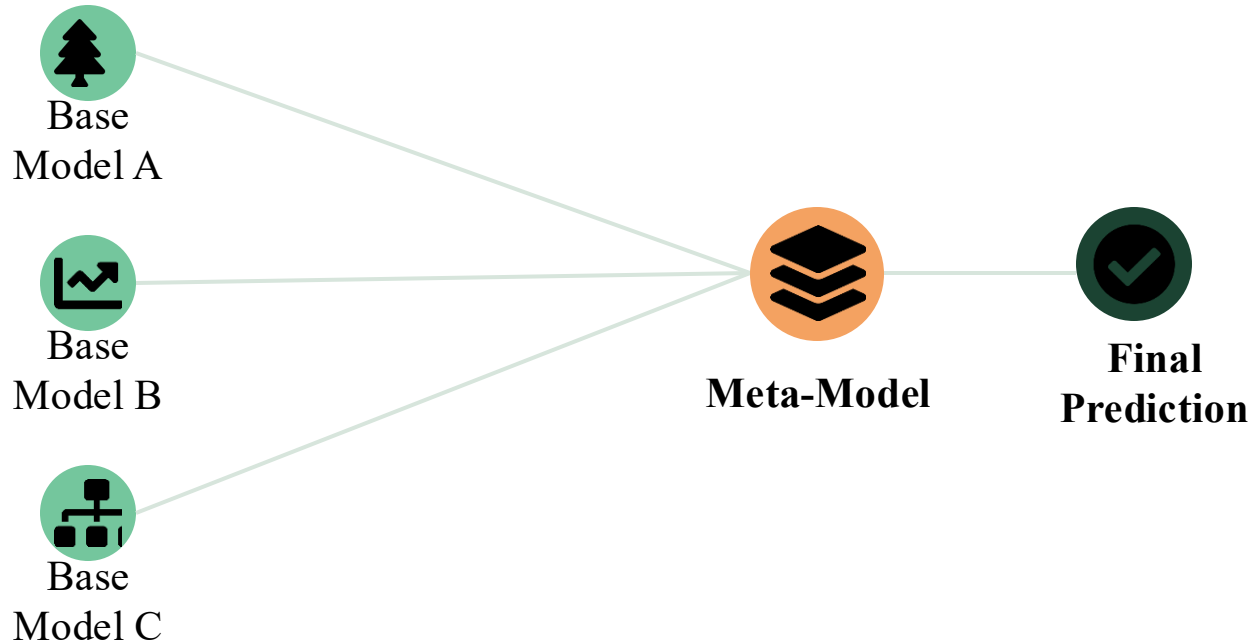


Key benefit: reduces bias by sequentially correcting errors — often yields very high accuracy, but can be sensitive to noisy data.



Stacking (Stacked Generalization)

Combines different types of models using a meta-model that learns how to best merge them.



- Different algorithms (e.g. decision tree, SVM, logistic regression) are trained as base models
- Their predictions become inputs to a second-level meta-model
- The meta-model learns the best way to combine the base predictions
- Often produces higher accuracy than any single base model or simple voting

Bagging vs Boosting vs Stacking

Three ensemble strategies, each suited to different situations.

Aspect	Bagging	Boosting	Stacking
Training Style	Parallel, independent	Sequential, corrective	Parallel base + meta layer
Main Goal	Reduce variance	Reduce bias	Combine model strengths
Base Models	Usually same type	Usually same type	Usually different types
Example Algorithm	Random Forest	AdaBoost, XGBoost	Stacked ensembles
Risk	Less prone to overfitting	Can overfit on noisy data	More complex to build

Advantages & Applications



Higher Accuracy

Combining models typically beats any single model



More Robust

Less sensitive to noise and outliers in the data



Reduces Overfitting

Especially true for bagging-based methods

Where it's used:

- Fraud & anomaly detection
- Medical diagnosis support
- Credit risk scoring
- Recommendation systems
- Kaggle-winning ML models

Thank You